

OS Protection Rings



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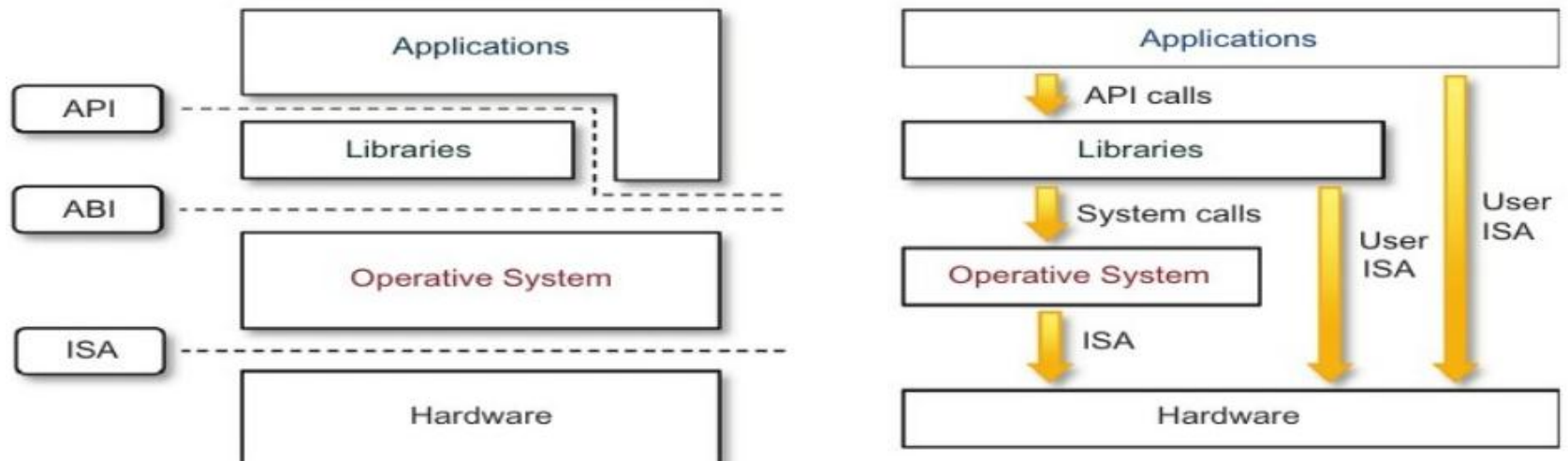
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Machine Reference Model



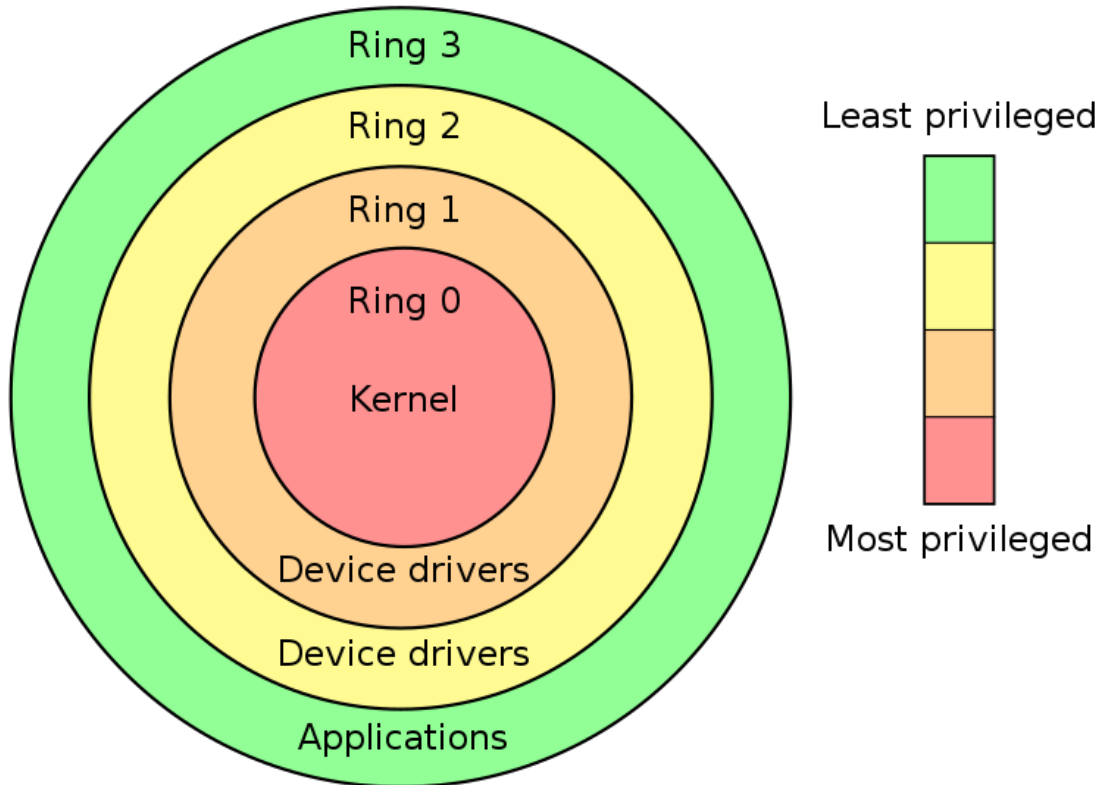
Machine Reference Model

- API, ABI and ISA are responsible for any operation to be performed in application level.
- High-level abstraction is converted into machine-level instructions to perform actual operations supported by the machine (processor).
- Physical resources, i.e. processors, registers and main memory capacities, are used to perform the operation at the H/W level of the CPU.
- Layered approach facilitates the development and implementation of computing systems and simplifies the implementation of multitasking and the coexistence of multiple executing environments.
- Layered model provides ways for implementing a minimal security model for managing and accessing shared resources.

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- Instruction set exposed by the hardware has been divided into different security classes that define who can operate with them.
- Instructions can be classified into
 - Privileged Instructions : Instructions that are executed under specific restrictions and are mostly used for sensitive operations, which expose (behavior-sensitive) or modify(control-sensitive) the privileged state.
 - Behavior-Sensitive : IO Operations
 - Control-Sensitive : Operations that alter the state of the CPU registers
 - Non Privileged : Non privileged instructions are those instructions that can be used without interfering with other tasks because they do not access shared resources (Example: all the floating, fixed-point, and arithmetic instructions)

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Thank You